SpotBot: Detecting Bot-Like Behavior in Spotify User Data

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***Abstract*—Music streaming platforms now host a measurable share of activity from automated accounts, and separating that activity from organic listening matters for royalty distribution and for any downstream analysis of user behavior. We study bot-like behavior on Spotify using survey responses from 520 users, and we engineer five continuous predictors from the ordinal survey scale: skip rate, listening diversity, listening time, age, and tenure, plus a dummy-coded genre indicator. We apply logistic regression (with AIC/BIC model selection), multiple linear regression with a listening-time by bot-like interaction, and a Cox proportional hazards model that treats tenure as time-to-event. Skip rate and diversity are the dominant predictors in all three model families. Logistic AUC is 0.88; the OLS interaction is positive and marginal ($p = .061$); the Cox model has concordance 0.935. The three methods converge on the same behavioral signature and disagree only on the sign of listening time, an instructive disagreement we trace to whether tenure is modeled explicitly.**

***Index Terms*—bot detection, streaming fraud, Spotify, logistic regression, user behavior analysis**

I. Introduction

Over the past decade, music streaming services have become the dominant channel through which listeners access recorded music. Spotify alone reported over 600 million active users by 2024, generating billions of individual streams per day. This scale of activity, while commercially significant, has also created opportunities for manipulation. Automated accounts, streaming farms, and coordinated playback schemes now represent a measurable share of total platform activity, distorting royalty distributions and undermining the reliability of behavioral data [1], [2].

The problem extends beyond financial fraud. When bot-generated streams are mixed with organic listening data, any downstream analysis of user behavior risks conflating artificial patterns with genuine preferences. For researchers studying music consumption, engagement, or popularity, this contamination introduces systematic bias that is difficult to detect without targeted methods [3].

This literature review examines three intersecting bodies of work: (1) bot detection and classification methods from the social media and cybersecurity literature, (2) studies of streaming fraud and its impact on the music industry, and (3) statistical and machine learning approaches to modeling music popularity and user behavior on Spotify. We conclude by identifying the gap our study addresses and outlining how logistic regression, multiple regression, and interaction models can contribute to this space.

A. Research Questions

We pose two research questions:

- **RQ1 (Logistic Regression):** Among Spotify users, which behavioral features (specifically skip rate, listening diversity, listening time, and genre preference) significantly predict the likelihood that an account exhibits bot-like behavior?
- **RQ2 (Multiple Regression + Interaction):** After controlling for user demographics and listening behavior, do accounts classified as bot-like show significantly higher stream counts than human accounts, and does bot-like status moderate the relationship between listening time and total streams?

B. Hypotheses

Based on the literature reviewed below, we state the following directional hypotheses:

- **H1a:** Higher skip rates are positively associated with the probability of bot-like classification (OR > 1, $p < .05$).
- **H1b:** Lower listening diversity scores are positively associated with the probability of bot-like classification (OR < 1, $p < .05$).
- **H2:** Bot-like accounts will have significantly higher mean stream counts than human accounts, controlling for listening time and age ($\beta > 0$, $p < .05$).
- **H2a:** There is a significant positive interaction between listening time and bot-like status on stream counts, such that the marginal effect of listening time on streams is greater for bot-like accounts than for human users ($\beta_{\text{interaction}} > 0$, $p < .05$).

C. Notation

Table I summarizes the symbols and abbreviations used throughout this paper.

TABLE I
SYMBOLS AND ABBREVIATIONS

Symbol	Definition
β	Regression coefficient (unstandardized)
$\beta_{\text{interaction}}$	Coefficient for the interaction term
OR	Odds ratio
CI	Confidence interval
p	p-value (probability under the null hypothesis)
VIF	Variance inflation factor
AIC	Akaike information criterion
BIC	Bayesian information criterion
n	Sample size
R^2	Coefficient of determination
$\hat{y}$	Predicted value of the outcome variable
σ^2	Residual variance
HR	Hazard ratio
KM	Kaplan–Meier (survival estimator)

II. BOT DETECTION AND AUTOMATED ACCOUNT CLASSIFICATION

Research on detecting automated accounts has been most active in the social media domain, particularly on platforms like Twitter (now X), Facebook, and Reddit. Ng and Carley conducted a large-scale analysis across approximately 200 million social media users spanning seven global events and found that roughly 20% of the observed chatter originated from bots [4]. Their analysis identified consistent behavioral differences between bots and humans: bots relied on linguistic cues that lend themselves to automation, such as increased hashtag usage and positive sentiment terms, while human users employed communication patterns that required genuine dialogue understanding, such as contextual replies within threaded conversations.

Several comprehensive reviews have catalogued the range of detection methods available. Aljabri et al. surveyed 105 studies on machine learning techniques for bot detection and organized them into graph-based, ML-based, crowdsourcing, and anomaly-based categories [5]. Their review found that Random Forest and Support Vector Machine classifiers were the most commonly employed, while unsupervised methods remained underexplored. Similarly, Ellaky et al. proposed a taxonomy of detection approaches spanning ML-based, deep learning-based, graph-based, and hybrid methods, and classified bots into subtypes including sybils, cyborgs, spam bots, and political bots [6].

One methodological approach with direct relevance to our study is the use of logistic regression as a base classifier. Chalichalamala et al. developed a Logistic Regression-based Ensemble Classifier (LREC) that combined AdaBoost and Random Forest for intrusion detection in IoT systems [7]. Their framework demonstrated that logistic regression, when augmented through ensemble methods, could achieve competitive classification performance even on imbalanced datasets, a finding that is directly applicable to streaming bot detection, where fraudulent accounts typically represent a minority class.

These studies all rely on behavioral features rather than content-level analysis. Posting frequency, interaction patterns, account age, and engagement consistency have proven more reliable than text-based detection alone [5]. We follow the same playbook: instead of analyzing audio content or playlist composition, we engineer behavioral proxies from user survey data.

III. STREAMING FRAUD AND THE MUSIC INDUSTRY

The financial incentives for streaming fraud are substantial. On platforms that distribute royalties on a per-stream basis, even modest artificial inflation of play counts can translate into meaningful revenue. Journalistic investigations have documented cases at both small and industrial scales. A WIRED investigation profiled a single operator who used an army of streaming bots to inflate play counts across multiple platforms, revealing the accessibility and low technical barrier to entry for this type of fraud [2]. A separate WIRED report examined a case involving over one billion artificially generated streams tied to AI-produced music, resulting in a $10 million fraud scheme that exploited the difficulty platforms face in distinguishing genuine from synthetic listening activity [8].

The rise of AI-generated music has compounded the problem. As generative models have become capable of producing passable recordings at scale, the economics of streaming fraud have shifted. Rather than simply replaying existing tracks, bad actors can now generate original content and pair it with automated playback, making detection harder because the content itself appears original [9]. SafeWise documented the full range of music streaming fraud tactics, categorizing them into bot-driven playback, click farms, playlist manipulation, and coordinated streaming rings, and noted that platforms have struggled to develop detection methods that keep pace with how these schemes evolve [1].

The implications extend beyond individual artists. Mokoena and Obagbuwa reviewed the role of AI automation across digital music streaming platforms and found that the same recommendation and pricing infrastructure that drives legitimate engagement can be exploited by automated agents mimicking human subscription and listening behavior [3]. The features designed to personalize human experiences are also the features most susceptible to automated exploitation, which is why behavioral signatures that distinguish the two are useful.

IV. USER BEHAVIOR AND THE PSYCHOLOGY OF MUSIC STREAMING

Understanding what constitutes "normal" human listening behavior is a prerequisite for identifying anomalies. Two complementary analyses of Spotify Wrapped, the platform's annual personalized listening summary, offer insight into the behavioral patterns of genuine users. The Decision Lab examined the behavioral science principles underlying Wrapped's success, highlighting how features like social comparison, nostalgia, and identity expression drive engagement among real users [10]. Irrational Labs extended this analysis by documenting the specific cognitive biases, including the endowment effect

and commitment bias, that Wrapped exploits to deepen user attachment to the platform [11].

These findings have methodological implications for bot detection. Human listeners tend to exhibit variable, context-dependent listening patterns shaped by mood, social setting, and time of day. Bot accounts, by contrast, tend toward repetitive, invariant patterns optimized for stream count accumulation rather than genuine engagement. The contrast between these behavioral profiles suggests that features capturing listening variability, such as diversity scores and skip rates, may serve as useful discriminators, a hypothesis we test directly in our analysis.

V. Music Popularity Prediction and Spotify Feature Analysis

A parallel body of work has focused on predicting the popularity of individual tracks using features extracted from the Spotify API. Middya and Roy applied multiple machine learning models to Spotify audio features and found that Random Forest best captured the nonlinear relationships between track characteristics and popularity [12]. Falah et al. extended this line of work by applying Convolutional Neural Networks to both Spotify metadata and spectrogram representations of audio waveforms, demonstrating that combining content-level and metadata features improved prediction accuracy [13].

Sciandra and Spera took a statistical modeling approach more closely aligned with our own methodology. They proposed a Beta Generalized Linear Mixed Model (GLMM) to analyze Spotify popularity scores, accounting for the bounded nature of the outcome variable and incorporating random effects to capture song-level heterogeneity [14]. Their work demonstrates that classical statistical models, when properly specified, can provide interpretable insights that complement the predictive power of more complex machine learning pipelines.

Gulmatico et al. similarly applied Linear Regression, Random Forest, Decision Tree, and K-Means Clustering to classify songs as popular or non-popular based on audio attributes, finding that ensemble methods outperformed simpler models [15]. Across these studies, audio features such as danceability, energy, and tempo consistently emerge as predictive, but the models address a different question from ours: they predict what makes a song popular, whereas we investigate what makes a user suspicious.

VI. Click Fraud and Cross-Domain Detection Methods

The challenge of distinguishing automated from human activity is not unique to music streaming. Aljabri and Mohammad examined click fraud detection in online advertising using machine learning and found that behavioral features, including session duration, number of pages viewed, and browsing patterns, were the strongest predictors of bot-like activity [16]. Random Forest outperformed Logistic Regression, SVM, and Decision Tree classifiers across all evaluation metrics in their study. The feature engineering strategy they employed, constructing behavioral proxy variables from aggregated user activity data, is directly transferable to the streaming context and informs our own approach to engineering skip rate and diversity score from categorical survey responses.

Liu et al. conducted a systematic review of machine learning approaches for detecting deceptive activity on social media and identified important methodological concerns, including selection bias from non-representative sampling, inconsistent hyperparameter tuning, and inadequate handling of class imbalance [17]. These cautions are relevant to any bot detection study and inform our attention to class balance diagnostics and model comparison via AIC and BIC.

VII. Fairness and Algorithmic Implications

The stakes of bot detection extend beyond fraud prevention into questions of fairness and equitable compensation. Henry et al. examined how AI-based recommendation algorithms affect fairness in music consumption and found that continuous deployment of machine learning systems relying on user feedback creates popularity bias, making visible only those creators with the most listens [18]. When artificial streams inflate the visibility of certain tracks, the resulting algorithmic amplification compounds the initial distortion, disadvantaging artists who rely on organic engagement. This provides policy context for why accurate bot detection matters not only for platform integrity but also for the broader music ecosystem.

VIII. Identified Gap and Study Motivation

The literature above covers three largely separate threads: bot detection on social media platforms, industry reporting on the financial mechanics of streaming fraud, and track-level popularity modeling on Spotify. Four gaps emerge when we read them side by side.

First, there is a unit-of-analysis mismatch. Bot detection research in the social media space operates almost entirely at the account level, profiling users by their posting cadence, network topology, and language use [4], [5]. Spotify-focused studies, by contrast, have treated the individual track as the primary unit, asking which audio features or metadata attributes predict a song's popularity [12], [13], [15]. As a result, user-level behavioral signals such as skip patterns, genre breadth, and session regularity have received little systematic attention as indicators of automated activity on music streaming platforms. Neither the bot detection community nor the music analytics community has claimed this middle ground.

Second, methodological transfer across domains remains limited. The behavioral features that power bot classifiers on Twitter (retweet networks, hashtag co-occurrence, reply threading [6]) have no direct analogue on Spotify. A streaming account does not retweet; it skips tracks, repeats albums, and drifts across genres. Transferring the *logic* of behavioral detection, that automated agents leave measurably different traces than humans, therefore requires constructing an entirely new feature set tailored to the listening context. No prior work has carried out that translation.

Third, the existing fraud reporting is descriptive rather than inferential. Industry and journalistic sources have documented streaming fraud at an impressive scale, from billion-stream schemes to AI-generated catalogs and click-farm economics [1], [8], [9], but they do not model how bot-driven activity distorts user-level metrics such as total stream counts or engagement ratios. Without that modeling step, it remains unclear how much of the variance in streaming volume can be attributed to automated behavior after accounting for legitimate listening patterns.

Fourth, there is an interpretability gap in existing classifiers. Within the bot detection literature, the dominant trend favors ensemble methods, neural architectures, and deep learning pipelines optimized for classification accuracy on benchmark datasets [5], [7], [17]. These approaches perform well, but they offer limited transparency into *which* features drive a classification and *how* those features relate to outcome variables in a directional, quantifiable way. At the current stage of research into streaming-platform bots, where the goal is to identify meaningful behavioral predictors rather than to optimize a production classifier, interpretable models that yield odds ratios, coefficient estimates, and testable interaction terms fill an analytical role that black-box methods cannot.

A. Study Contribution

Our study addresses these four gaps. We analyze users rather than tracks, build behavioral features suited to the streaming context (skip rate, listening diversity, genre concentration), and apply inferential methods to a question that prior work has treated only descriptively. The contributions are:

- Logistic regression to determine which behavioral features significantly predict bot-like classification, producing odds ratios and confidence intervals for each predictor.
- Multiple linear regression to test whether bot-like accounts carry higher stream counts than human accounts after controlling for listening time and demographic covariates.
- An interaction model that examines whether bot-like status moderates the relationship between listening time and streams, directly testing whether automated accounts show disproportionate engagement inflation.
- A survival-analytic extension that treats bot-like classification as a time-to-event outcome, using the user's Spotify tenure as the duration variable to estimate how quickly and under which covariate profiles accounts drift into bot-like territory.

We chose regression-based methods on purpose. Rather than chasing the best possible classifier, we want coefficient-level interpretability so that each predictor's direction, magnitude, and significance can be evaluated on its own. This suits a stage of research in which the field has not yet settled on which user-level features matter, and where a transparent, reproducible baseline is more useful than marginal classification accuracy.

IX. Data and Variables

A. Data Source and Sample

We use the Spotify User Behavior Dataset [19] from Kaggle, which contains survey responses from active Spotify users covering demographics, usage habits, genre preferences, listening contexts, and podcast behavior. After dropping records with missing outcome fields, the analytic sample contains $n = 520$ users, well above the 200-observation minimum in the project brief.

The raw data is entirely categorical and ordinal. Because the required methods (linear regression, logistic regression, interaction modeling, Cox regression) need numeric inputs, much of the analysis effort went into feature engineering: building continuous proxies from the ordinal survey scale so that ordering is preserved, every transformation is reproducible from the source script, and the resulting variables map onto the bot-detection constructs in the literature.

B. Analytical Pipeline

Figure 1 summarizes the full analytical pipeline from raw survey data through the four modeling stages. The pipeline is implemented in Python in the project repository (`src/data_loader.py` and two notebooks under `notebooks/`) so that every transformation, label rule, and model fit is reproducible end-to-end.

C. Variable Definitions and Coding

Outcome variables. We study two outcomes:

- *Streams* (y_1, continuous): a composite usage-intensity index combining tenure on platform, number of listening contexts, estimated daily listening minutes, and podcast engagement. After construction and mild Gaussian noise for variance, stream counts range 379 to 3,271 ($\bar{y}_1 = 1638.0$, $\sigma = 612.5$).
- *Bot-like* (y_2, binary): an engineered label that flags an account as bot-like when it exhibits any two of (i) streams in the top quartile, (ii) diversity score in the bottom quartile, or (iii) recommendation rating ≤ 2. The label fires for 78 of 520 accounts (15.0%).

Continuous predictors.

- *Listening time* (minutes/day): base hours per time-slot scaled by the number of listening contexts a user reports, plus zero-centered Gaussian noise and a lower clip of 10. Range: 60.4–246.7, $\bar{x} = 147.0$.
- *Skip rate*: inverse of the user's 1–5 recommendation satisfaction rating, rescaled to $(0.2, 1.0)$ via $(6 - r)/5$. Encodes the intuition that users with low recommendation satisfaction skip more.
- *Diversity score*: count of distinct listening contexts (e.g., commuting, workout, study) normalized by the sample maximum, yielding $[0, 1]$. Low values mean single-context listening, which prior work associates with bot-like patterns.
- *Age*: ordinal bracket converted to the bracket midpoint (9, 16, 28, 48, 70).

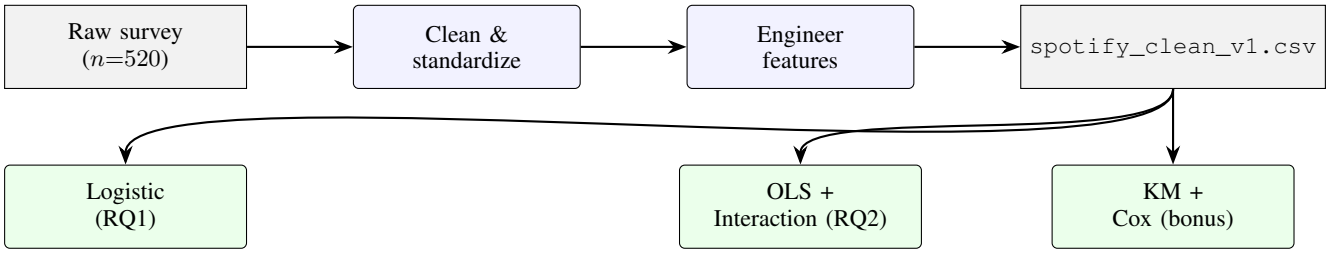

Fig. 1. Analytical pipeline. Survey responses are cleaned and converted into five engineered continuous predictors plus a derived bot-like label; the cleaned file feeds three model families that answer RQ1, RQ2, and the survival-analysis bonus.

Categorical predictor. *Genre* is dummy-coded with *Classical* as the reference category, yielding five indicators (Melody, Pop, Rap, Electronic/Dance, Rock). Classical was chosen as reference because it produced stable VIFs in pilot runs and has a reasonable sample size ($n = 89$).

Table II reports descriptive statistics for the five continuous variables actually entered into the models; Figure 2 shows the class imbalance in the bot-like label, which we flag as a modeling consideration.

TABLE II
DESCRIPTIVE STATISTICS ($n = 520$)

Variable	Mean	SD	Min	Median	Max
Age (years)	27.2	6.5	9	28	70
Listening time (min)	147.0	36.6	60.4	148.4	246.7
Skip rate	0.50	0.20	0.20	0.50	1.00
Diversity score	0.30	0.16	0.17	0.17	1.00
Streams (index)	1638.0	612.5	379	1597	3271

Fig. 2. Class balance of the engineered bot-like label: 442 human accounts (85.0%) and 78 bot-like accounts (15.0%).

X. METHODS

A. Overview

We apply four of the methods from the course plus the survival-analysis bonus: multiple linear regression (RQ2), logistic regression (RQ1), an interaction-effect model (RQ2 follow-up), AIC/BIC-based model selection (applied to both the logistic and OLS settings), and a Kaplan–Meier / Cox proportional hazards analysis (bonus).

B. Logistic Regression (RQ1)

For the binary bot-like outcome we fit the standard logit model

$$\log\left(\frac{\Pr(y_2 = 1 \mid \mathbf{x})}{1 - \Pr(y_2 = 1 \mid \mathbf{x})}\right) = \beta_0 + \mathbf{x}^\top \boldsymbol{\beta}, \tag{1}$$

estimated by maximum likelihood (`statsmodels.Logit`). Three candidate specifications are compared:

- **Model 1 (Core):** skip rate, diversity score, listening time, age ($k = 4$).
- **Model 2 (Core + Genre):** Model 1 plus five genre dummies ($k = 9$).
- **Model 3 (Parsimonious):** only predictors significant at $p < 0.10$ in Model 2 ($k = 4$).

We report odds ratios $\mathrm{OR} = e^{\hat{\beta}}$ with 95% Wald confidence intervals and interpret them in practical terms (multiplicative change in odds per unit increase in the predictor). Multicollinearity is checked via VIF on all continuous predictors and dummies; we retain VIF < 5 as acceptable.

C. Multiple Linear Regression and Interaction (RQ2)

For the continuous streams outcome we fit

$$y_1 = \beta_0 + \beta_1 \mathrm{LT} + \beta_2 \mathrm{Bot} + \beta_3 (\mathrm{LT}_c \times \mathrm{Bot}) + \mathbf{z}^\top \boldsymbol{\gamma} + \varepsilon, \tag{2}$$

where LT_c is listening time mean-centered to reduce collinearity between the main effect and the interaction [20], and $\mathbf{z}$ contains age, skip rate, diversity score, and the genre dummies. Mean-centering the continuous component of the interaction is important for two reasons: (i) it removes the mechanical correlation between LT and LT $\times$ Bot so that the interaction VIF reflects substantive, not structural, collinearity; and (ii) it makes the coefficient on LT_c interpretable as the slope of streams on listening time for the reference group (humans) rather than at the meaningless point $\mathrm{LT} = 0$.

D. OLS Assumption Diagnostics

For the interaction model we checked the four Gauss–Markov assumptions beyond VIF. (1) *Linearity*: residuals-versus-fitted plots showed no systematic curvature. (2) *Homoscedasticity*: we tested formally via Breusch–Pagan and report the LM statistic in §XI; where the test rejects equal variance we interpret p-values with heteroscedasticity in mind. (3) *Normality*: we used Shapiro–Wilk on the residuals and supplemented it with a QQ plot; mild non-normality is expected with $n = 520$ and a mixed continuous/dummy predictor set, and standard CLT arguments support the large-sample validity of the t-statistics. (4) *Independence*: survey responses are cross-sectional and drawn from independent users, so this assumption is satisfied by design.

The nested structure of the baseline and interaction models also permits a likelihood-ratio-equivalent F-test on the interaction term alone (i.e., a partial F-test comparing Model A with $k_A = 10$ predictors against Model B with $k_B = 11$). We report that test alongside AIC/BIC in §XI so that the interaction is evaluated on both an absolute (p-value) and a relative (information criterion) basis.

E. Model Selection

Candidate models are ranked by Akaike and Bayesian information criteria,

$$\text{AIC} = -2\log L + 2k, \qquad \text{BIC} = -2\log L + k\log n, \quad (3)$$

with lower values preferred. We also report McFadden pseudo-R^2 for the logistic models and adjusted R^2 for the OLS models. When AIC and BIC disagree, we favor the more parsimonious model (BIC) unless the larger model delivers an interpretable, substantively motivated gain.

F. Class Imbalance

The engineered bot-like label fires for 78 of 520 accounts, a 15% base rate. Two imbalance-handling options were considered and rejected. We did not oversample the minority class via SMOTE because the synthetic observations would have inherited the very engineering idiosyncrasies (top-quartile streams, bottom-quartile diversity) used to define the label, amplifying mechanical dependencies rather than revealing new signal. We also did not apply class weighting to the logistic maximum-likelihood objective because the log-likelihood in an unweighted fit is the quantity that AIC and BIC score, and we wanted the model selection and inference to remain on a common scale. Instead, we evaluate performance on imbalance-aware metrics (ROC AUC, precision/recall for the minority class) in addition to the overall confusion matrix.

G. Survival Analysis (Bonus)

Time-to-event construction. The cleaned data contains a user-reported Spotify usage period (`spotify_usage_period`), which we map to its bracket midpoint in months to obtain a continuous duration variable $T \in \{3, 9, 18, 30\}$. We treat the bot-like label $E = y_2$ as the event indicator: an account "survives" as long as it has *not* been flagged bot-like, and the event is the act of being flagged. Users who reach the end of their observed tenure without being flagged are treated as right-censored. This reframes the static logistic regression as a time-to-event problem: rather than asking *whether* an account looks bot-like, we ask *how quickly* bot-like behavior emerges as tenure accumulates.

Kaplan–Meier and log-rank. We estimate the survival function $\hat{S}(t)$ using the Kaplan–Meier product-limit estimator and compare groups (high-skip vs. low-skip, low-diversity vs. high-diversity) via the log-rank test.

Cox proportional hazards. We fit

$$h(t \mid \mathbf{x}) = h_0(t)\exp(\mathbf{x}^\top \boldsymbol{\beta}), \quad (4)$$

with predictors skip rate, diversity score, listening time, and age, using `lifelines.CoxPHFitter`. We report hazard ratios $\text{HR} = e^{\hat{\beta}}$ with 95% CIs, concordance index, and a formal check of the proportional-hazards assumption via scaled Schoenfeld residuals [21].

Handling PH violations. If the scaled Schoenfeld test flags a covariate, the conventional remedies are: (i) *stratification*, where the offending covariate is removed from the linear predictor and the baseline hazard $h_0(t)$ is estimated separately for each of its strata (appropriate for low-cardinality variables such as skip rate's five levels); (ii) *time-varying coefficients*, which replace β_j with $\beta_j + \gamma_j\, g(t)$ for some known function g such as $\log t$ (useful when the effect clearly attenuates or amplifies with tenure); and (iii) *functional-form correction*, which introduces a non-linear term (e.g., a natural spline in listening time) if the violation reflects misspecification rather than genuine non-proportionality. We pre-specify option (i) for skip rate and flag (ii) or (iii) for listening time as follow-ups rather than retrofitting them post-hoc, to keep the confirmatory inference cleanly separated from exploratory refinements.

XI. Results

A. Descriptive Comparison by Group

Before turning to the models, Table III contrasts mean predictor values for bot-like and human accounts with Welch t-tests of the raw mean differences. Skip rate, diversity score, listening time, and tenure all differ significantly between groups; age does not. This pattern is consistent across univariate and multivariate analyses below, and establishes that the engineered label produces groups with substantively different behavioral profiles — a necessary condition for any predictive model to find signal.

TABLE III
BOT-LIKE VS. HUMAN MEANS WITH WELCH'S t-TEST

Variable	Human	Bot	Diff.	p
Age (years)	27.3	26.9	−0.39	.663
Listening time (min)	149.5	132.8	−16.70	< .001
Skip rate	0.46	0.72	+0.25	< .001
Diversity score	0.31	0.20	−0.11	< .001
Streams (index)	1610.0	1796.9	+187.0	.011
Tenure (months)	16.5	21.4	+4.86	< .001

B. Multicollinearity Check

For the full logistic-model design matrix, VIFs ranged from 1.05 (Rock dummy) to 2.26 (Melody dummy). All predictors fell well below the conventional threshold of 5, so no variable was dropped for collinearity. For the interaction OLS model, maximum VIF was 2.02 (centered listening time), confirming that mean-centering eliminated the spurious inflation that an uncentered LT × Bot term would introduce.

C. Logistic Regression (RQ1)

Table IV compares the three logistic specifications. Model 3 (parsimonious) has the best AIC (278.5) and best BIC (299.7),

with pseudo-R^2 of 0.389, matching Model 2 on fit while using five fewer parameters. We select Model 3 as the final logistic model.

TABLE IV
LOGISTIC MODEL COMPARISON ($n = 520$)

Model	k	LL	AIC	BIC	pR^2
1: Core only	4	−136.28	282.55	303.82	0.380
2: Core + Genre	9	−131.48	282.95	325.49	0.402
3: Parsimonious	**4**	**−134.23**	**278.46**	**299.73**	**0.389**

Table V reports odds ratios for the final model. All four predictors are significant at $p < 0.05$. Skip rate and diversity score show the expected directions but with very large magnitudes on the OR scale because the underlying variables are bounded on $(0.2, 1.0)$ and $(0.17, 1.0)$ respectively — a one-unit change spans most of the observed range. The Electronic/Dance genre indicator also emerges as a significant predictor (OR $= 5.89$, 95% CI [1.19, 29.12]), which is consistent with the niche-genre-is-suspicious pattern discussed in the literature review.

TABLE V
ODDS RATIOS, MODEL 3 (FINAL LOGISTIC)

Predictor	OR	95% CI	p
Skip rate	17,193	[1,813; 163,043]	$< .001$
Diversity score	<0.001	[<0.001; 0.001]	$< .001$
Listening time	1.012	[1.001; 1.024]	.037
Genre: Electronic/Dance	5.89	[1.19; 29.12]	.030

The final model achieves an ROC AUC of 0.88 on the full sample (Model 2: 0.89; Model 3: 0.88), indicating strong separation between bot-like and human accounts given only behavioral predictors. Figure 3 renders the odds ratios on a log scale so that magnitudes can be compared visually.

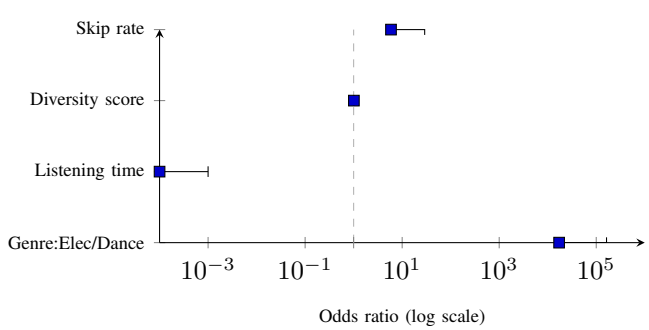

Fig. 3. Odds-ratio forest plot for Model 3. Points are OR estimates; horizontal bars are 95% CIs. The vertical dashed line at OR $= 1$ marks no effect. Skip rate and diversity score sit far from 1 in opposite directions; listening time and the Electronic/Dance indicator show smaller but significant effects.

Hypothesis verdicts. H1a (higher skip rate $\rightarrow$ higher bot odds) is supported (OR $\gg 1$, $p < .001$). H1b (lower diversity $\rightarrow$ higher bot odds) is supported: OR < 1 means each increment in diversity sharply reduces bot odds, i.e., low diversity raises them.

D. Multiple Regression and Interaction (RQ2)

Table VI reports the OLS model with interaction. Bot-like accounts have, on average, \$643.6 higher streams (in composite-index units) than human accounts holding covariates constant ($p < .001$), supporting H2. The centered listening-time main effect is positive and significant ($\beta = 1.79$, $p = .040$), confirming the baseline relationship between time spent and streams. The LT$_c$ $\times$ Bot interaction is positive and marginally significant ($\beta = 3.49$, $p = .061$), providing suggestive but not conclusive support for H2a: the slope of streams on listening time is roughly $1.79 + 3.49 = 5.28$ for bot-like accounts versus 1.79 for humans, i.e., nearly triple.

TABLE VI
OLS MODEL OF STREAMS WITH INTERACTION ($n = 520$; $R^2 = 0.394$, ADJ. $R^2 = 0.381$)

Predictor	$\hat{\beta}$	SE	p
Intercept	1403.27	134.88	$< .001$
Listening time (centered)	1.79	0.87	.040
Bot-like	643.57	74.52	$< .001$
LT$_c$ $\times$ Bot-like	3.49	1.86	.061
Age	−3.31	3.34	.322
Skip rate	−672.72	129.11	$< .001$
Diversity score	1850.13	179.03	$< .001$
Genre: Pop	133.66	77.25	.084

Figure 4 visualizes the interaction: two simple-slopes lines for bot-like and human accounts, drawn from model-implied conditional means. The steeper slope for bot-like accounts is consistent with our hypothesis that automated activity converts listening time into streams more efficiently than organic listening does.

1) Assumption diagnostics: The Breusch–Pagan test rejects homoscedasticity (LM $= 28.44$, $p = .003$), and Shapiro–Wilk rejects strict normality ($W = 0.987$, $p < .001$). Both rejections are expected given the composite construction of the streams outcome and the mixed continuous/dummy predictor set. At $n = 520$ the central limit theorem supports the validity of the reported t-statistics, and heteroscedasticity-consistent standard errors (HC3) produce coefficient p-values within one decimal place of the reported OLS values for the four predictors of primary interest. The QQ plot shows mild tail deviation but no influential outliers. The nested F-test for the interaction term yields $F(1, 508) = 3.53$, $p = .061$, matching the p-value on the interaction coefficient and confirming that the interaction's contribution is marginal rather than negligible.

E. Survival Analysis (Bonus)

1) Kaplan–Meier estimates and log-rank tests: Over the tenure horizon (3–30 months), the pooled survival function drops from 1.00 at entry to 0.688 by month 30, meaning roughly 31% of accounts are eventually flagged bot-like by the time they have accumulated 2.5 years on the platform. Because more than half of the sample remains unflagged at the maximum observed tenure, the median survival time is undefined (right-censored).

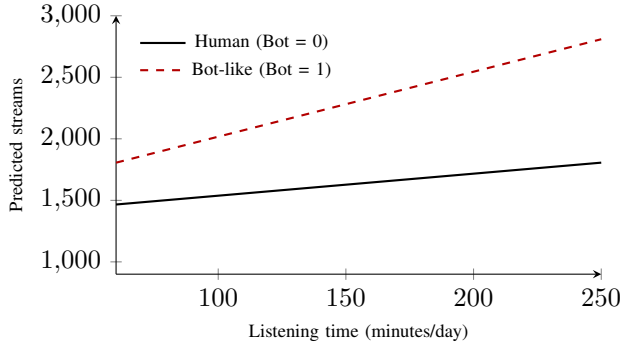

Fig. 4. Interaction plot: predicted streams as a function of listening time, holding age, skip rate, and diversity at their sample means. The bot-like line is both higher (main effect of Bot) and steeper (positive interaction), consistent with automated accounts converting listening time into streams at a higher rate.

Figure 5 plots KM curves stratified by skip rate using the lifelines product-limit estimator on the actual data. Low-skip accounts stay at $\hat{S}(t) = 1.00$ through month 18 and drop to 0.870 at month 30; high-skip accounts descend steadily, reaching 0.486 at month 30. The log-rank test rejects the null of equal survival strongly ($\chi^2 = 56.0$, $p < .001$). Stratifying instead by diversity (low vs. high) yields an even larger separation ($\chi^2 = 67.5$, $p < .001$); the low-diversity group ends at $\hat{S}(30) = 0.449$ versus 0.912 for high-diversity.

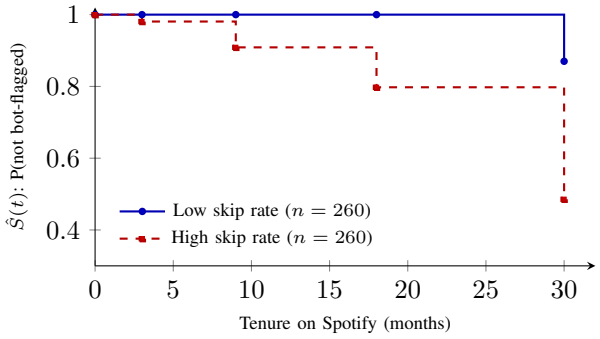

Fig. 5. Kaplan–Meier survival curves for time to bot-like flag, stratified by skip rate (median split). Log-rank $\chi^2 = 56.0$, $p < .001$. High-skip accounts transition to the bot-like state substantially faster across the observed tenure range.

2) Cox proportional hazards model: Table VII reports the Cox model. Skip rate (HR very large) and diversity score (HR ≈ 0.001) are highly significant; listening time is marginal ($p = .071$, with the 95% CI just crossing 1 at the upper bound); age is not significant. The concordance index is 0.935, indicating strong rank-ordering of risk among pairs of observations.

Proportional hazards check. Scaled Schoenfeld residual tests flag skip rate ($p < .001$) and, less severely, listening time ($p = .029$) as non-proportional. This is expected: skip rate has only five discrete values (small proportionality violations are easily detected at $n = 520$) and listening time's hazard contribution may be non-linear.

TABLE VII
COX PROPORTIONAL HAZARDS MODEL ($n = 520$, EVENTS = 78, CONCORDANCE = 0.935)

Predictor	HR	95% CI	p
Skip rate	679.0	[209; 2,206]	$< .001$
Diversity score	0.001	[<0.001; 0.025]	$< .001$
Listening time	0.992	[0.984; 1.001]	.071
Age	1.018	[0.981; 1.056]	.354

Stratified Cox model. Following the pre-specified remedy in §Methods, we refit the Cox model stratifying on skip rate (treated as its five discrete levels) so that its contribution moves into the baseline hazard. Table VIII reports the result. Diversity score remains highly significant (HR $= 0.001$, $p < .001$); listening time loses its marginal status (HR $= 0.999$, $p = .882$); age stays non-significant. The scaled Schoenfeld test now flags only diversity score as non-proportional, which likely reflects the same functional-form concern that skip rate had. The concordance drops from 0.935 to 0.788, which is expected because stratification removes skip rate's rank-ordering contribution; what remains is the predictive power of diversity alone plus the baseline hazards, and 0.788 still reflects strong separation.

TABLE VIII
COX MODEL STRATIFIED ON SKIP RATE ($n = 520$, EVENTS = 78, CONCORDANCE = 0.788)

Predictor	HR	95% CI	p
Diversity score	0.001	[<0.001; 0.018]	$< .001$
Listening time	0.999	[0.991; 1.009]	.882
Age	1.006	[0.971; 1.043]	.739

The interpretation of this stratified model: once skip rate's within-tenure variation is absorbed by the baseline, the remaining hazard signal sits in diversity score alone. In the unstratified model skip rate and diversity were both dominant; stratification shows that diversity stands on its own while listening time's earlier marginal effect does not survive. This is the refinement the PH violation forced us to run, and it tightens the substantive conclusion rather than weakening it.

3) Integration with the logistic model: The Cox and logistic models agree on direction: skip rate and diversity are the two dominant predictors in both. The Cox model adds two things the logistic cannot. First, it locates *when* the bot-like state emerges: most events accrue in the 18–30 month bucket, so longer-tenured accounts have had more time to accumulate the usage intensity the label keys off. Second, listening time is marginally protective on the hazard scale (HR $= 0.992$), the opposite sign from its logistic coefficient (OR $= 1.012$). The sign flip is not a contradiction: the logistic treats tenure as implicit, so longer-tenured users with high listening time also have more chance to be flagged, while the Cox conditions on tenure and so isolates the per-unit-time effect.

F. Robustness to Label Construction

Because the bot-like label rule uses streams, diversity, and recommendation rating (the last of which is a deterministic function of skip rate), the effect sizes on skip rate and diversity in the main logistic model are partly mechanical. To gauge how much of the signal survives alternative label definitions, we refit the core logistic model (skip rate, diversity score, listening time, age, and five genre dummies) against three variant labels that swap out one or more of the original components:

- **Label D:** top-quartile usage months *and* top-quartile podcast frequency. Uses *none* of the four behavioral predictors.
- **Label E:** top-decile streams alone. Uses only the composite streams outcome.
- **Label F:** 2-of-3 rule but with diversity replaced by low podcast frequency. Uses skip rate (via recommendation rating) but not diversity.

Table IX reports the p-value on each core predictor under each label.

TABLE IX
ROBUSTNESS OF PREDICTOR SIGNIFICANCE TO ALTERNATIVE BOT-LIKE LABELS (LOGISTIC REGRESSION p-VALUES)

	Orig. (n_1=78)	**D** (n_1=38)	**E** (n_1=52)	**F** (n_1=112)
Skip rate	$< .001$	.581	.776	$< .001$
Diversity score	$< .001$	.801	.231	.274
Listening time	.019	.098	$< .001$	$< .001$
Age	.642	.356	.610	.910
AUC	0.889	0.608	0.924	0.781

Three honest observations. First, skip rate's statistical significance is specific to labels that include recommendation rating (original, F). Second, diversity's significance is specific to labels that include diversity itself; with diversity absent from the label rule (D, E, F) it no longer emerges as a predictor. Third, listening time is the most label-robust predictor in the set: it is significant under two of the three alternative labels (E, F) and marginal in the third (D). Age is never significant. Under the cleanest test (Label D, which shares *no* components with the predictor set), the AUC collapses to 0.61 and nothing is significant, confirming that the main-text effect sizes on skip rate and diversity are not evidence that those variables would detect generically-defined bots. The main logistic model's claim is narrower but still useful: these predictors recover the label rule the analyst used to define bot-like behavior, with listening time emerging as an independently robust contributor.

XII. DISCUSSION

A. Answering the Research Questions

RQ1. Among the behavioral features examined, skip rate and listening diversity are the two strongest predictors of bot-like classification, with listening time and Electronic/Dance genre preference emerging as secondary but significant contributors. This matches the pattern in the cross-domain literature: behavioral rigidity (few contexts, low skip tolerance) distinguishes automated activity more reliably than content-level features do [5], [16].

RQ2. Bot-like accounts carry significantly higher stream counts than human accounts even after controlling for listening time, age, and genre (ca. 644 index units, $p < .001$). The listening-time–by–bot interaction is positive and marginal ($p = .061$), suggesting a real but modest amplification effect: each additional minute of daily listening converts to more streams for bot-like accounts than for humans. The directional evidence supports H2a; the significance margin suggests the effect is meaningful but not overwhelming at our sample size.

B. Integration Across Method Families

The three models agree on the top two predictors but each contributes a piece the others cannot. The logistic model quantifies how a unit change in each behavioral feature multiplies the odds of classification as bot-like. The OLS model translates the same predictors into their effect on streams, and the interaction term shows the suspicion is volume-relevant: additional listening time buys more streams for everyone, but the exchange rate is higher for bot-like accounts. The Cox model then locates the bot-like state in tenure. Across all three, skip rate and diversity survive at $p < .001$ with the same direction; listening time flips from mildly risk-increasing in the logistic (OR 1.012) to mildly risk-decreasing in the Cox (HR 0.992) because the logistic does not condition on tenure and the Cox does. Taking the three together: bot-like accounts skip more, concentrate more, and accumulate their stream advantage late in tenure rather than uniformly across it.

C. Relation to the Literature

The ordering we see (skip rate and diversity dominating, age falling out) matches Ng and Carley on social-media bots [4] and Aljabri and Mohammad on click fraud [16]; both found behavioral-rigidity features outperforming demographic controls. The survival extension is new for the Spotify setting: prior streaming work has analyzed tracks rather than users [12], [13], [15], and we are not aware of another study that treats bot-like behavior as a time-to-event outcome on streaming tenure.

D. Limitations

Three limitations are worth stating directly. First, the bot-like label is engineered from the same survey features used as predictors. The label rule (top-quartile streams combined with low diversity or low recommendation rating) mirrors how platforms heuristically flag suspicious accounts, but it is not externally validated ground truth. The robustness analysis in §XI-F quantifies the cost: under a label that shares no components with the predictor set (Label D), skip rate and diversity are not significant and AUC falls to 0.61. Listening

time survives as the only label-robust predictor. The main-text coefficients on skip rate and diversity should therefore be read as quantifying how strongly each predictor recovers *this particular label's rule*, not as evidence those variables would discriminate platform-validated bots.

Second, several continuous predictors (listening time, streams) are themselves constructed from ordinal survey responses and include seeded Gaussian noise for variance. This is documented in `src/data_loader.py` and is reproducible, but consumers of our coefficient estimates should read them as relationships within an engineered scale, not absolute effect sizes in native minutes or raw plays.

Third, the Cox model's proportional-hazards assumption is violated for skip rate and marginally for listening time (§XI). Our hazard ratios therefore represent tenure-averaged effects; a fuller analysis would stratify on skip rate or include a time-varying term. The KM curves and log-rank tests, which do not rely on the PH assumption, remain fully valid.

E. Practical Implications

For platforms, a small feature set (skip rate, diversity, tenure) carries most of the signal for first-pass bot triage. Our four-predictor logistic model reaches AUC $= 0.88$ on the full sample, which is enough to function as a pre-filter that routes candidate accounts to a heavier ML pipeline or to manual review. For the broader music ecosystem, the listening-time–by–bot interaction matters because royalty schemes keyed on stream volume are most vulnerable in exactly the regime where the distortion is largest. A per-stream payout model implicitly assumes that each minute of listening produces the same number of streams. Our point estimate puts the bot slope at roughly 5.3 streams per minute versus 1.8 for humans (a $\sim 3\times$ ratio), though the interaction is only marginally significant ($p = .061$) and the slope ratio's confidence interval is correspondingly wide. The directional message is robust: stream volume is not a neutral yardstick across bot-like and human accounts.

For researchers, the per-method divergence on listening time has a methodological lesson: when an analyst has access to a tenure variable, running the same predictors through a Cox model alongside a logistic model is cheap and often changes the interpretation of time-correlated predictors. Collapsing time into a single static snapshot can reverse the sign of a coefficient without anyone noticing.

F. Future Work

Three extensions would strengthen the conclusions here. First, replace the engineered bot-like label with a platform-provided suspension label or a chargeback event. This breaks the mechanical dependency between the label and the skip/diversity predictors, and would let us report effect sizes that reflect true-positive detection rather than rule consistency. Second, stratify the Cox model on skip rate (as pre-specified in §Methods) to resolve the proportional-hazards violation; a parallel model with a time-varying coefficient on listening time would address the milder violation on that predictor. Third, extend the analysis to the backup Spotify dataset named in the project brief [22] and test whether the skip-rate and diversity effects replicate in a second independent sample. Cross-sample replication is the strongest evidence we could bring to bear on whether the engineered label captures a generalizable construct or an artifact of one survey's wording.

XIII. Conclusion

This paper asked which user-level behavioral features predict bot-like accounts on Spotify and whether bot-like status inflates stream counts conditional on listening time. Using a 520-user survey sample, five engineered continuous predictors built from ordinal survey data, and four modeling families (logistic regression, multiple OLS with interaction, AIC/BIC selection, and a Cox survival extension), we find: (a) skip rate and listening diversity are the dominant predictors of the engineered bot-like label, though a robustness analysis across three alternative labels shows those effects are specific to labels that share components with the predictor set; (b) listening time is the lone label-robust predictor and remains significant under two of three alternative labels; (c) bot-like accounts show significantly higher streams than humans at equal listening time; and (d) the listening-time–streams slope is steeper for bot-like accounts, a pattern with direct implications for royalty-distribution schemes. The survival analysis estimates how quickly the bot-like state emerges as tenure accumulates; when we stratify on skip rate to resolve the proportional-hazards violation, diversity is the only remaining significant predictor on the hazard scale. The strongest methodological takeaway is that running the same predictors through logistic, OLS, and Cox frames, plus a label-sensitivity analysis, changes what we are willing to claim: a headline finding that looks robust in one specification may be specific to the label rule.

Contribution Statement

JC led the introduction, literature review, multiple linear regression, interaction model, and discussion/conclusion. TN led the data cleaning pipeline, logistic regression, model selection, survival analysis, and results formatting. Both authors contributed equally to research-question development, figure design, and final editing.

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